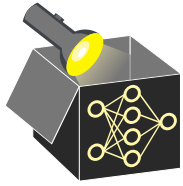
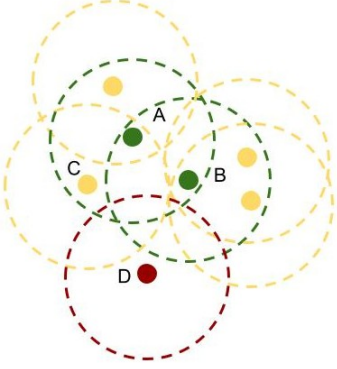
Interpretable Machine Learning



Local Explanations:
Increasing Trust in Explanations

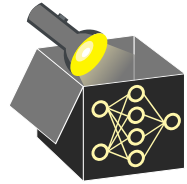


Learning goals

- Understand the aspects that undermine users' trust in an explanation
- Learn diagnostic tools that could increase trust

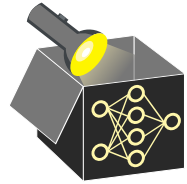
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- Local explanations should not only make a model interpretable but also reveal if the model is trustworthy

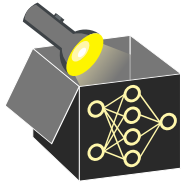


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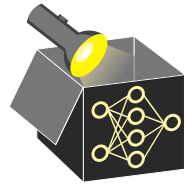


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     - Failure case: generation is based on inputs in areas where the model was trained with little or no training data (extrapolation)

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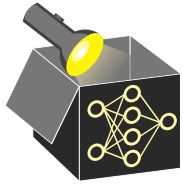


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     - Expectation: similar explanations for similar data points with similar predictions
     - However, multiple sources of uncertainty exist

     ⇝ measure how robust an IML method is to small changes in the input data or parameters

     ⇝ Is an observation out-of-distribution?

MOTIVATION & IMPORTANT PROPERTIES



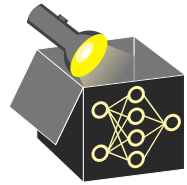
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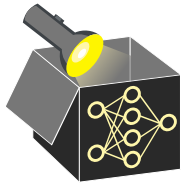
     $\rightsquigarrow$ Is an observation out-of-distribution?
- Failing in one of these $\rightsquigarrow$ undermining users’ trust in the explanations
  $\rightsquigarrow$ undermining trust in the model

OUT-OF-DISTRIBUTION (OOD) DETECTION

- Models are unreliable in areas with little data support
  $\rightsquigarrow$ explanations from local explanation methods are unreliable

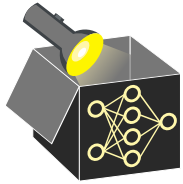


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- For local explanation methods, the following components could be out-of-distribution (OOD):
  - The data for LIME's surrogate model
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  - Shapley value's permuted obs. to calculate the marginal contribs
  - ICE curves grid data points

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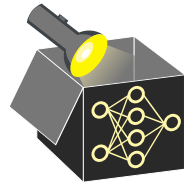


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- Two very simple and intuitive approaches
  - Classifier for out-of-distribution
  - Clustering
- More complicated also possible, e.g., variational autoencoders
  - "Daxberger et al." 2020

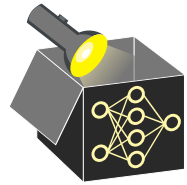
OOD DETECTION: OOD-CLASSIFIER

- Problem: we have only in-distribution data
- Idea: Hallucinate new (ood) data by randomly sampling data points

⇝ Learn a binary classifier to distinguish between the origins of the data



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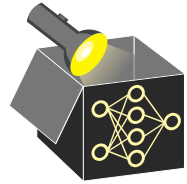
⇝ Learn a binary classifier to distinguish between the origins of the data

- Study whether an explanation approach can be fooled ▸ "Dylan Slack et al." 2020
  - Hide bias in the true (deployed) model, but use an unbiased model for all out-of-distribution samples

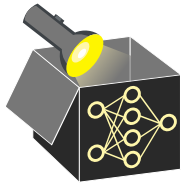
⇝ Important way to diagnose an explanation approach

OOD DETECTION: CLUSTERING VIA DBSCAN

- DBSCAN is a data clustering algorithm ▸ "Martin Ester et al." 1996
  (Density-Based Spatial Clustering of Applications with Noise)



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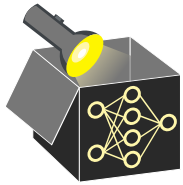


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- For this method, we define an ϵ-neighborhood:
  Given a dataset $X = \{\mathbf{x}^{(i)}\}_{i=1}^{n}$, an ϵ-neighborhood for $\mathbf{x} \in \mathcal{X}$ is defined as

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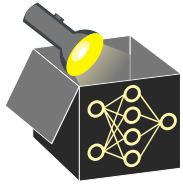


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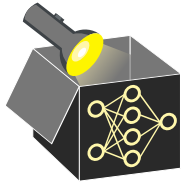


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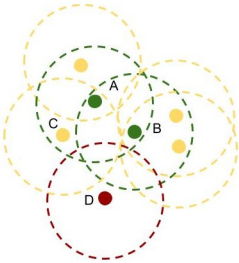


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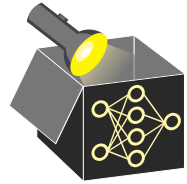
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- Noise points
  - Are not within $\mathcal{N}_\epsilon(\mathbf{x})$
  - Not part of any cluster

OUT-OF-DISTRIBUTION DETECTION

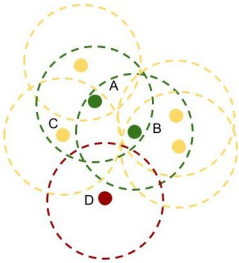


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- Green points A and B are core points and form one cluster since they lie in each others neighborhood, all yellow points are border points of this cluster

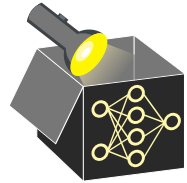


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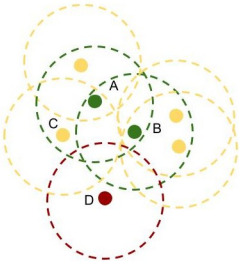


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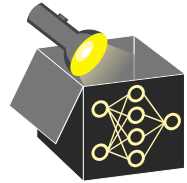


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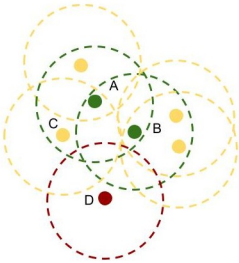


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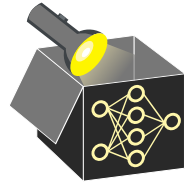


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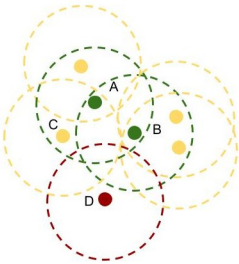


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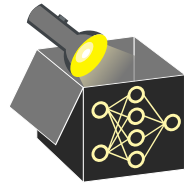


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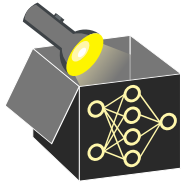
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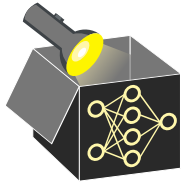
- Disadvantages:
  - Depending on the distance metric $d(\cdot)$, DBSCAN could suffer from the "curse of dimensionality"
  - The choice of ϵ and m is not clear a-priori

ROBUSTNESS

- Differentiate between different kinds of uncertainty:
  1. **Explanation uncertainty**: Explanation changes if we repeat the process, e.g., the explanation could differ depending on which subset of data we use for the expl. method and which hyperparams

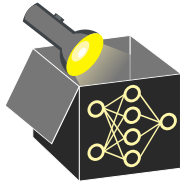


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     $\rightsquigarrow$ are ML models non-robust, e.g., because of training on noisy data?

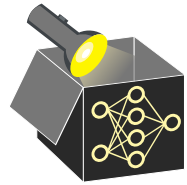
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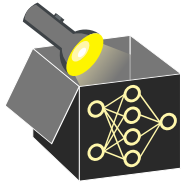
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- We focus on explanation uncertainty
  - Even with the same model and same (or similar) data points, we can receive different explanations

ROBUSTNESS MEASURE FOR LIME AND SHAP

- Objective: Similar explanations for similar inputs (in a neighborhood)



ROBUSTNESS MEASURE FOR LIME AND SHAP

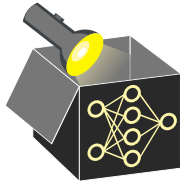


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- For LIME and SHAP, notion of stability based on **locally Lipschitz continuity**
  - "Alvarez-Melis and Jaakkola" 2018:

  An explanation method $g : \mathcal{X} \rightarrow \mathbb{R}^m$ is locally Lipschitz if
  - for every $\mathbf{x}_0 \in \mathcal{X}$ there exist $\delta > 0$ and $\omega \in \mathbb{R}$
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  Note that, for LIME, g returns the m coefficients of the surrogate model

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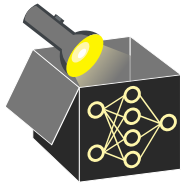
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- According to this, we can quantify the robustness of explanation models in terms of ω:

  $\rightsquigarrow$ The closer ω is to 0, the more robust our explanation method is

ROBUSTNESS MEASURE FOR LIME AND SHAP



- Objective: Similar explanations for similar inputs (in a neighborhood)
- For LIME and SHAP, notion of stability based on **locally Lipschitz continuity** ▸ "Alvarez-Melis and Jaakkola" 2018:

  An explanation method $g : \mathcal{X} \to \mathbb{R}^m$ is locally Lipschitz if
  - for every $\mathbf{x}_0 \in \mathcal{X}$ there exist $\delta > 0$ and $\omega \in \mathbb{R}$
  - such that $||\mathbf{x} - \mathbf{x}_0|| < \delta$ implies $||g(\mathbf{x}) - g(\mathbf{x}_0)|| < \omega||\mathbf{x} - \mathbf{x}_0||$

  Note that, for LIME, g returns the m coefficients of the surrogate model
- According to this, we can quantify the robustness of explanation models in terms of ω:

  $\rightsquigarrow$ The closer ω is to 0, the more robust our explanation method is
- ω is rarely known a-priori but it could be estimated as follows:

$$\hat{\omega}_X(\mathbf{x}) \in \underset{\mathbf{x}^{(i)} \in \mathcal{N}_\epsilon(\mathbf{x})}{\arg\max} \frac{||g(\mathbf{x}) - g(\mathbf{x}^{(i)})||_2}{d(\mathbf{x}, \mathbf{x}^{(i)})}$$

where $\mathcal{N}_\epsilon(\mathbf{x})$ is the ϵ-neighborhood of $\mathbf{x}$